EV Charging Station Analytics

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1. Data Quality Impact

Issue:

- Inconsistent vehicle_id created duplicate categories. Differences like capitalization, prefixes, and naming conventions.

Fix:

- Standardized IDs and mapped to clean names

Results:

- There were fewer categories therefore cleaner patterns in charging demand

If Skipped:

- This could cause misleading results like overstimulated vehicle diversity and weak conclusions.

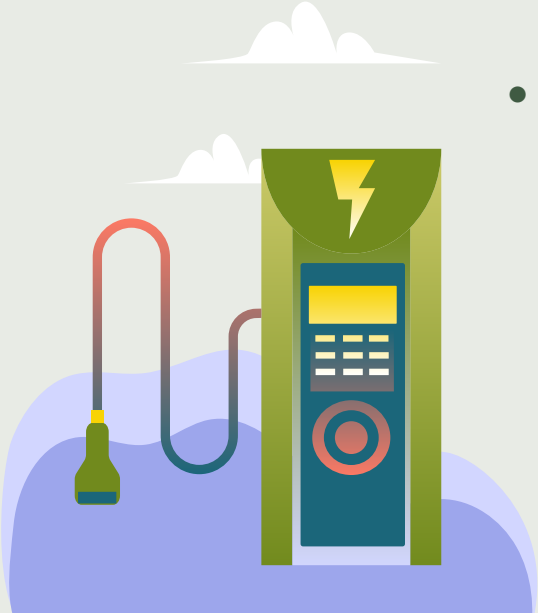
2. Cloud Pipeline Experience

Challenge:

- GCS paths and folder structure were confusing
- Authentication differences across environments

Improvements:

- Define paths upfront and validate uploads early



Comparison FIndings:

- Local - simple but not shared
- Cloud - harder to set up but better for collaboration and reproducability

3. ETL Trade-offs

Decision:

- Converted negative kwh_delevered to positive using .abs()

Alternative:

- Dropped the rows
- Flagged as errors

Trade off:

- Kept dataset complete and easier to analyze

But

- Assumed negatives were mistakes (may lose nuance)

4. Pipeline Trust

Fragile Parts:

- Hardcoded GCS file paths
- Vehicle mapping since if there is not match for a vehicle ID they would turn into a missing value.

Risk:

- Pipeline could fail silently or produce incomplete data

Fix:

- Added file checks and handled unmapped values

5. Team Collaboration

Work Split:

- Sunhanee - GCS/cloud setup, authentication, and debugging
- Emily - Data cleaning pipeline (date parsing, fix cost column, and vehicle consolidation)
- Tina and Sydney - Analytical questions, visualizations, and pipeline integration

Another week:

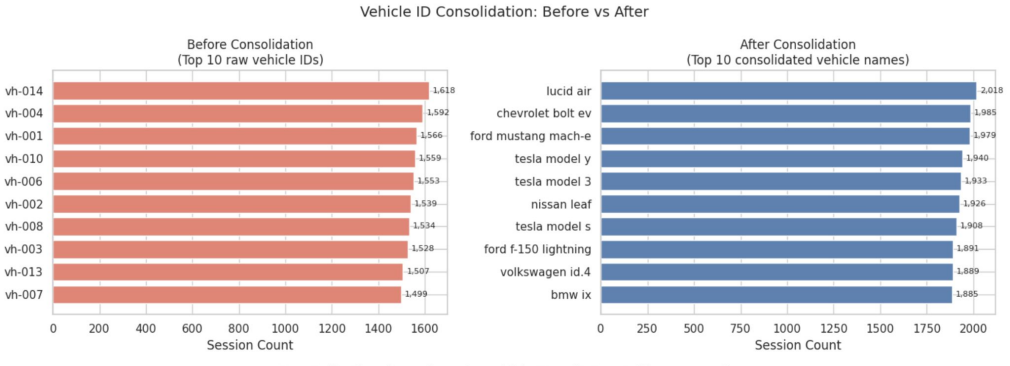
- More targeted validation of cleaned data
- Catch edge cases
- Improve automation so pipeline runs smoothly without manual steps

Valuable Skills:

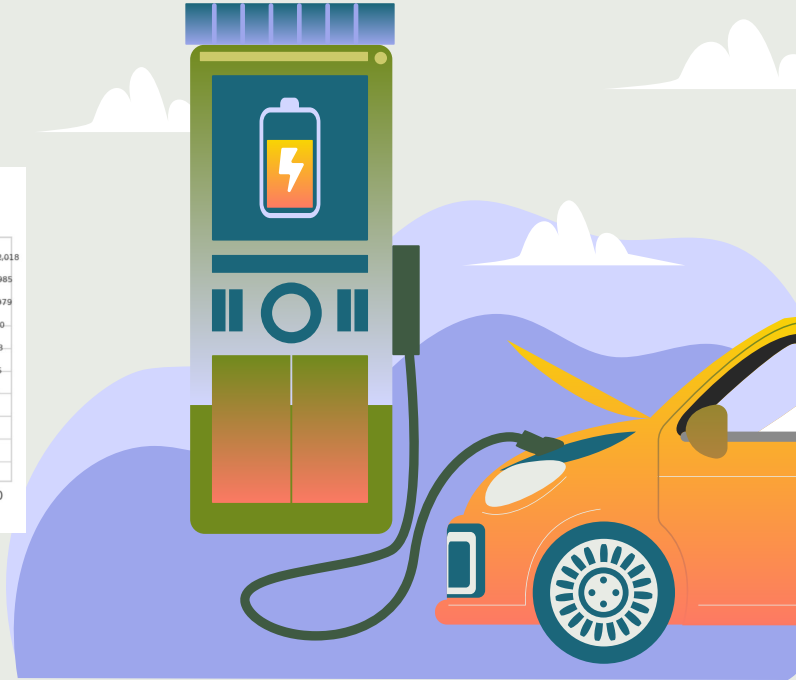
- Sunhanee - Cloud setup and ensuring pipeline works
- Emily - Data cleaning
- Tina and Sydney - Analysis and validation

What We Learned

- We learned the importance of structuring data properly before performing any analysis. Things like organized tables and ensuring consistent IDs

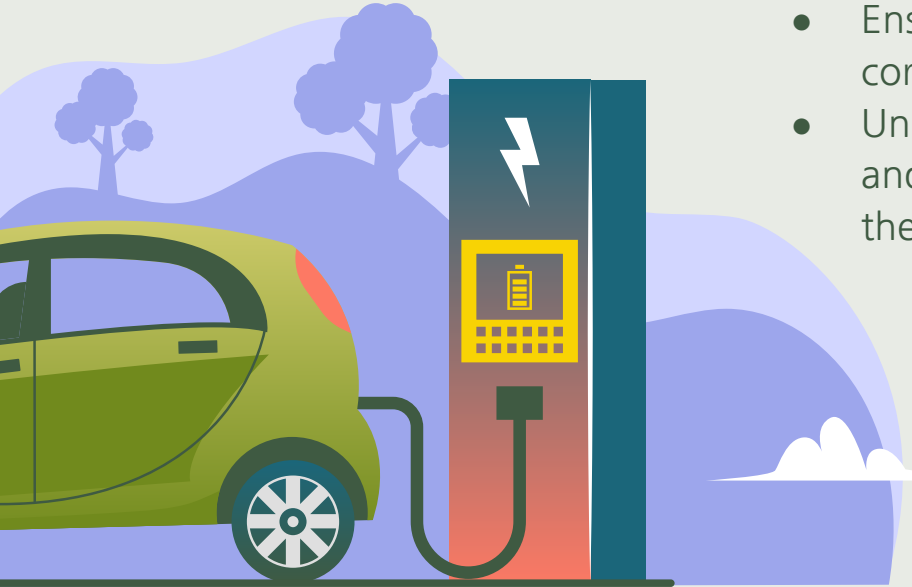


- The maximum amount of a vehicle ID increased showing a mismatch between vehicle IDs before.



What We Learned

- Working with real world data shows that datasets are rarely clean and ready for analysis. There could be slight inconsistencies that could greatly affect the results.
- This reinforced that data preparation and transformation are just as important as the final analysis, since errors at this stage can propagate through the entire project.

Challenges

- One major challenge was dealing with inconsistent formats across the dataset. This made joining, grouping, and analyzing data difficult.
- Ensuring that our transformations were correct without accidentally introducing errors.
- Understanding how the different tables relate and making sure data stayed consistent across them.